CT UNIVERSITY, LUDHIANA (PUNJAB)

SCHOOL OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(ARTIFICIAL INTELLIGENCE & DATA SCIENCE)

DAILY DIARY / INTERNSHIP TRAINING LOG BOOK

OFFLINE SHORT-TERM INDUSTRIAL TRAINING ON AI AUTOMATION & AGENT ENGINEERING
(18 JUNE 2026 – 14 AUGUST 2026)

CAPSTONE PROJECT:

DermoScan: AI-Powered Cutaneous Triage & Clinical Morphology Workstation

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DEPLOYMENT & REPOSITORY:
Live URL: https://dermo-scan-nesn.vercel.app/
GitHub: https://github.com/vijayesh26/Dermo-Scan-

18 June 2026

1. Objectives of the Day: To understand the internship schedule, the Datavalley training structure, and the basics of AI Automation and Agentic AI.

2. Summary of Work Done / Tasks Performed: Attended the orientation at Datavalley, learned about the internship program and its modules, and studied what AI Automation and Agentic AI mean, with real-world uses such as business process automation and customer support.

3. Skills/Concepts Learned: The difference between AI models, AI agents, and agentic AI systems, and an overview of Large Language Models (LLMs) and Generative AI.

4. Challenges Faced: Telling apart a simple AI model from an autonomous agent that plans and uses tools.

5. Solutions/Support Received: Followed the trainer's explanations and compared each concept with everyday applications.

6. Conclusion/Reflection: Gained a clear starting point for the internship and understood the topics to be covered in the coming weeks.

19 June 2026

1. Objectives of the Day: To learn the fundamentals of Prompt Engineering for working with LLMs such as ChatGPT, Claude, and Gemini.

2. Summary of Work Done / Tasks Performed: Practised writing prompts with clear instructions, context, and constraints, and compared the quality of responses for different prompt styles.

3. Skills/Concepts Learned: Clear instruction design, context management, and the effect of prompt wording on output quality.

4. Challenges Faced: Vague prompts produced inconsistent answers.

5. Solutions/Support Received: Added context, the expected format, and examples to the prompts and re-ran them to compare the outputs.

6. Conclusion/Reflection: Understood that the quality of the prompt directly decides the quality of the AI output.

22 June 2026

1. Objectives of the Day: To apply advanced prompting techniques for more reliable AI outputs.

2. Summary of Work Done / Tasks Performed: Worked with role-based prompting (researcher, analyst, support assistant), step-by-step instructions, and few-shot examples, and tested how each changed the responses.

3. Skills/Concepts Learned: Role-based prompting, structured prompting, and improving prompts through iteration.

4. Challenges Faced: Keeping long prompts focused without losing important instructions.

5. Solutions/Support Received: Broke long prompts into labelled parts and refined them over several attempts.

6. Conclusion/Reflection: Improved my ability to get consistent, well-structured results from AI models.

23 June 2026

1. Objectives of the Day: To understand the Natural Language Processing (NLP) concepts used in AI systems.

2. Summary of Work Done / Tasks Performed: Studied tokenization, text understanding, and intent recognition, and reviewed how NLP supports chatbots and document analysis.

3. Skills/Concepts Learned: Basics of NLP, tokens, and how text is prepared for AI processing.

4. Challenges Faced: Understanding how a model interprets context and meaning rather than just individual words.

5. Solutions/Support Received: Worked through short text examples with the trainer to see how text is processed step by step.

6. Conclusion/Reflection: Built a base for the upcoming work on structured outputs and retrieval.

24 June 2026

1. Objectives of the Day: To learn how AI models return structured outputs and call external tools.

2. Summary of Work Done / Tasks Performed: Practised getting responses in JSON format and studied function calling, where a model triggers a tool or API instead of only writing text.

3. Skills/Concepts Learned: JSON structure, structured outputs for system integration, and the basics of function calling.

4. Challenges Faced: Small formatting mistakes made some JSON outputs unusable.

5. Solutions/Support Received: Specified the exact JSON fields in the prompt and validated each output.

6. Conclusion/Reflection: Understood how structured outputs let AI connect with other applications.

25 June 2026

1. Objectives of the Day: To apply the earlier concepts by building the mini project, an AI Resume Analyzer.

2. Summary of Work Done / Tasks Performed: Built an AI-powered tool that reads a resume, assesses skills, experience, and qualifications, and returns structured JSON feedback with improvement suggestions.

3. Skills/Concepts Learned: Combining prompt engineering, structured outputs, and LLMs in one working solution.

4. Challenges Faced: Getting the analysis in a consistent format for different resumes.

5. Solutions/Support Received: Refined the prompt and the JSON structure after testing with several sample resumes.

6. Conclusion/Reflection: Completed the first mini project and gained confidence in building AI-based tools.

29 June 2026

1. Objectives of the Day: To understand workflow automation and get started with N8N.

2. Summary of Work Done / Tasks Performed: Studied what workflow automation is, explored the N8N interface and its nodes, and built a simple workflow connecting two applications.

3. Skills/Concepts Learned: Workflow automation concepts, nodes, and visual workflow building in N8N.

4. Challenges Faced: Understanding how data passes from one node to the next.

5. Solutions/Support Received: Inspected the output of each node and followed the trainer's demo workflow.

6. Conclusion/Reflection: Became comfortable with the N8N workspace and the idea of no-code automation.

30 June 2026

1. Objectives of the Day: To learn event-driven automation using triggers and actions.

2. Summary of Work Done / Tasks Performed: Created workflows started by events such as a new email or a form submission, and explored Zapier as another no-code automation platform.

3. Skills/Concepts Learned: Triggers, actions, event-driven workflows, and a comparison of N8N and Zapier.

4. Challenges Faced: Choosing the right trigger for each task.

5. Solutions/Support Received: Tested each trigger with sample events and checked the workflow results.

6. Conclusion/Reflection: Understood how events start automated processes.

1 July 2026

1. Objectives of the Day: To connect applications using APIs and webhooks.

2. Summary of Work Done / Tasks Performed: Studied how APIs let AI systems communicate with external services, set up a webhook to start a workflow, and sent test requests.

3. Skills/Concepts Learned: APIs, webhooks, HTTP requests and responses, and the basics of API authentication.

4. Challenges Faced: Handling errors when a request failed or returned unexpected data.

5. Solutions/Support Received: Checked the request details and response codes and corrected the configuration.

6. Conclusion/Reflection: Learned how real-time integrations link AI workflows to other tools.

2 July 2026

1. Objectives of the Day: To build an AI-powered Gmail auto-reply workflow.

2. Summary of Work Done / Tasks Performed: Created a workflow that detects a new email, sends its content to an LLM, and drafts a suitable reply automatically.

3. Skills/Concepts Learned: Integrating Gmail, an LLM, and automation nodes in one workflow.

4. Challenges Faced: Getting replies with the right tone and content.

5. Solutions/Support Received: Improved the prompt given to the LLM and tested with different sample emails.

6. Conclusion/Reflection: Completed a working email auto-reply automation.

3 July 2026

1. Objectives of the Day: To extend the auto-reply workflow into a full AI Email Automation Bot.

2. Summary of Work Done / Tasks Performed: Added email categorization, conditional routing, and reply generation, then tested and optimized the workflow.

3. Skills/Concepts Learned: Conditional logic in workflows, email classification, and workflow testing and optimization.

4. Challenges Faced: Routing different types of email to the correct path.

5. Solutions/Support Received: Added decision nodes and retested with sample emails of each type.

6. Conclusion/Reflection: Finished the email automation project and learned how AI reduces manual communication work.

6 July 2026

1. Objectives of the Day: To understand multi-agent systems and the CrewAI framework.

2. Summary of Work Done / Tasks Performed: Studied why several specialised agents work better than one for complex tasks, and explored the basic structure of a CrewAI project.

3. Skills/Concepts Learned: Multi-agent architecture and the basics of CrewAI.

4. Challenges Faced: Understanding how agents share the work.

5. Solutions/Support Received: Went through the trainer's example crew step by step.

6. Conclusion/Reflection: Understood the idea behind collaborative AI agents.

7 July 2026

1. Objectives of the Day: To define agents with clear roles, goals, and tools.

2. Summary of Work Done / Tasks Performed: Created agents such as Researcher, Planner, and Writer, gave each a goal, and attached tools such as web search.

3. Skills/Concepts Learned: Agent roles, goals, backstory, and tool usage.

4. Challenges Faced: Writing role and goal descriptions specific enough to give good results.

5. Solutions/Support Received: Rewrote the role descriptions and compared how the agents behaved.

6. Conclusion/Reflection: Learned how well-defined roles improve agent performance.

8 July 2026

1. Objectives of the Day: To understand agent memory, collaboration, and the difference between AI models and AI agents.

2. Summary of Work Done / Tasks Performed: Enabled memory for agents, assigned tasks between them, and compared a plain LLM response with an agent that plans and uses tools.

3. Skills/Concepts Learned: Agent memory, task delegation, agent communication, and models versus agents.

4. Challenges Faced: Keeping context consistent between agents.

5. Solutions/Support Received: Reviewed the task outputs passed between agents and adjusted the task descriptions.

6. Conclusion/Reflection: Understood how memory and communication make agents effective.

9 July 2026

1. Objectives of the Day: To build an AI Travel Planner using a team of agents.

2. Summary of Work Done / Tasks Performed: Designed agents for destination research, itinerary planning, budget estimation, and final report writing, and connected them in a sequential workflow.

3. Skills/Concepts Learned: Designing and orchestrating a multi-agent workflow.

4. Challenges Faced: Coordinating the order of tasks between agents.

5. Solutions/Support Received: Defined clear inputs and expected outputs for each task.

6. Conclusion/Reflection: Built the first working version of the travel planner team.

10 July 2026

1. Objectives of the Day: To test the travel planner and begin learning Retrieval-Augmented Generation (RAG).

2. Summary of Work Done / Tasks Performed: Tested the planner with different destinations and budgets, then studied how RAG combines document retrieval with LLM responses.

3. Skills/Concepts Learned: Agent testing, the RAG concept, and how retrieval reduces hallucinations.

4. Challenges Faced: Budget estimates were inconsistent in some outputs.

5. Solutions/Support Received: Adjusted the budget agent's instructions and re-tested.

6. Conclusion/Reflection: Completed the travel planner project and started the RAG module.

13 July 2026

1. Objectives of the Day: To learn the types of RAG and how embeddings represent meaning.

2. Summary of Work Done / Tasks Performed: Studied Basic, Advanced, Agentic, and Hybrid RAG, and how text is converted into embeddings.

3. Skills/Concepts Learned: RAG architectures and the idea of embeddings.

4. Challenges Faced: Understanding how numbers can capture the meaning of text.

5. Solutions/Support Received: Used simple examples to compare similar and different sentences.

6. Conclusion/Reflection: Understood how RAG architectures differ and where each one is used.

14 July 2026

1. Objectives of the Day: To understand vector databases and semantic search.

2. Summary of Work Done / Tasks Performed: Studied how embeddings are stored in vector databases and searched by similarity, and tried retrieving relevant text for sample questions.

3. Skills/Concepts Learned: Vector databases, similarity search, and semantic versus keyword search.

4. Challenges Faced: Understanding why meaning-based search returns different results from keyword search.

5. Solutions/Support Received: Ran the same query with both approaches and compared the results.

6. Conclusion/Reflection: Understood that retrieval is the core of document-based AI applications.

15 July 2026

1. Objectives of the Day: To build AI pipelines using the no-code platform Stack AI.

2. Summary of Work Done / Tasks Performed: Used the visual workflow builder to connect an input, an LLM, and a knowledge source into a working pipeline.

3. Skills/Concepts Learned: No-code AI pipeline design, and LLM and knowledge base integration in Stack AI.

4. Challenges Faced: Connecting the knowledge source correctly to the model.

5. Solutions/Support Received: Followed the platform's templates and tested the pipeline repeatedly.

6. Conclusion/Reflection: Built a first AI pipeline without writing much code.

16 July 2026

1. Objectives of the Day: To learn the basics of deploying AI applications and start the PDF Chatbot project.

2. Summary of Work Done / Tasks Performed: Studied workflow testing, API configuration, access management, and monitoring, and began designing document upload and retrieval for the PDF Chatbot.

3. Skills/Concepts Learned: Deployment concepts and designing a document ingestion workflow.

4. Challenges Faced: Planning how a document moves from upload to answer.

5. Solutions/Support Received: Drew the workflow in steps before building it.

6. Conclusion/Reflection: Completed phase 1 of the PDF Chatbot design.

17 July 2026

1. Objectives of the Day: To integrate document retrieval into the PDF Chatbot.

2. Summary of Work Done / Tasks Performed: Processed PDF content, created embeddings, stored them, and connected retrieval to the LLM so that answers come from the document.

3. Skills/Concepts Learned: Document processing, embedding generation, and retrieval integration.

4. Challenges Faced: Some answers did not match the document content.

5. Solutions/Support Received: Adjusted how the text was split and tested again with several questions.

6. Conclusion/Reflection: The chatbot began answering from the uploaded document.

20 July 2026

1. Objectives of the Day: To improve the functionality and accuracy of the PDF Chatbot.

2. Summary of Work Done / Tasks Performed: Improved the prompts, handled questions that fall outside the document, and tested the chatbot on multiple PDFs.

3. Skills/Concepts Learned: Grounded responses and improving retrieval accuracy.

4. Challenges Faced: The chatbot sometimes gave answers that the document did not support.

5. Solutions/Support Received: Instructed the model to answer only from retrieved content and to say when the information is unavailable.

6. Conclusion/Reflection: Completed the PDF Chatbot project.

21 July 2026

1. Objectives of the Day: To understand autonomous AI agents.

2. Summary of Work Done / Tasks Performed: Studied goal-oriented behaviour, decision making, tool use, and multi-step execution, and built a simple goal-driven agent.

3. Skills/Concepts Learned: Autonomous agent design and tool-based decision making.

4. Challenges Faced: Keeping the agent focused on its goal.

5. Solutions/Support Received: Defined the goal and limits clearly and reviewed the agent's steps.

6. Conclusion/Reflection: Built a goal-driven agent that completes tasks with little guidance.

22 July 2026

1. Objectives of the Day: To study how multiple agents collaborate and communicate.

2. Summary of Work Done / Tasks Performed: Designed the flow of information between a coordinator agent and specialist agents, including task delegation and result sharing.

3. Skills/Concepts Learned: Agent communication flow, coordination, and task delegation.

4. Challenges Faced: Avoiding repeated or conflicting work between agents.

5. Solutions/Support Received: Gave each agent a separate responsibility and passed results in a fixed order.

6. Conclusion/Reflection: Understood how communication design affects a multi-agent system.

23 July 2026

1. Objectives of the Day: To learn how AI agents plan and execute tasks.

2. Summary of Work Done / Tasks Performed: Practised splitting a large goal into smaller tasks, setting priorities, and tracking progress so the plan can change when needed.

3. Skills/Concepts Learned: Task decomposition, priority management, and execution monitoring.

4. Challenges Faced: Judging the right order of tasks.

5. Solutions/Support Received: Reviewed the plan after each step and corrected the sequence.

6. Conclusion/Reflection: Learned how planning lets agents handle complex objectives.

24 July 2026

1. Objectives of the Day: To build a Multi-Agent Research Assistant.

2. Summary of Work Done / Tasks Performed: Created a research agent to collect information, an analysis agent to evaluate the findings, and a report agent to prepare a summary.

3. Skills/Concepts Learned: Research automation, data analysis by agents, and automated report generation.

4. Challenges Faced: Making the outputs of the three agents fit together.

5. Solutions/Support Received: Defined a shared format for what each agent passes on.

6. Conclusion/Reflection: Completed the Multi-Agent Research Assistant mini project.

27 July 2026

1. Objectives of the Day: To understand AI customer support systems.

2. Summary of Work Done / Tasks Performed: Studied automated query resolution, intelligent chatbots, and ticket categorization and routing, and designed a sample support workflow.

3. Skills/Concepts Learned: Customer support automation and the use of NLP and LLMs in support systems.

4. Challenges Faced: Handling queries that the system cannot answer.

5. Solutions/Support Received: Added a fallback step that passes such queries to a human.

6. Conclusion/Reflection: Learned how AI improves response time and support quality.

28 July 2026

1. Objectives of the Day: To learn the basics of AI voice assistants.

2. Summary of Work Done / Tasks Performed: Studied the pipeline of speech recognition, language understanding, response generation, and text-to-speech.

3. Skills/Concepts Learned: Speech-to-text, intent understanding, and text-to-speech.

4. Challenges Faced: Understanding how the stages connect to each other.

5. Solutions/Support Received: Reviewed the flow with a diagram and examples.

6. Conclusion/Reflection: Gained a basic understanding of voice-enabled AI.

29 July 2026

1. Objectives of the Day: To apply AI agents to business tasks and improve workflow efficiency.

2. Summary of Work Done / Tasks Performed: Automated tasks such as scheduling, report preparation, and email drafting, and reviewed how to reduce steps and errors in the workflows.

3. Skills/Concepts Learned: Business process automation and workflow optimization.

4. Challenges Faced: Finding which steps can be safely automated.

5. Solutions/Support Received: Measured each step and removed the unnecessary ones.

6. Conclusion/Reflection: Understood where AI agents add the most value in business processes.

30 July 2026

1. Objectives of the Day: To understand end-to-end AI system design and monitoring.

2. Summary of Work Done / Tasks Performed: Studied the data, processing, AI, automation, and user interaction layers, and how dashboards track performance, workflow runs, and errors.

3. Skills/Concepts Learned: AI system architecture, monitoring, error tracking, and logging.

4. Challenges Faced: Seeing how all the layers fit together.

5. Solutions/Support Received: Drew a layered diagram of a complete AI solution.

6. Conclusion/Reflection: Learned the main parts of a complete AI system.

31 July 2026

1. Objectives of the Day: To connect AI with everyday tools and finalise the capstone project topic.

2. Summary of Work Done / Tasks Performed: Integrated AI workflows with Gmail, Google Drive, and calendar tools, and finalised the capstone project: DermoScan: AI-Powered Cutaneous Triage & Clinical Morphology Workstation.

3. Skills/Concepts Learned: AI integration with productivity tools and scoping a project.

4. Challenges Faced: Deciding the project scope within the time available.

5. Solutions/Support Received: Listed the modules and kept the ones achievable in the remaining days.

6. Conclusion/Reflection: Completed the training phase and became ready to begin the capstone project.

3 August 2026 – Capstone Project Inception & Clinical Scoping

1. Objectives of the Day: To conceptualize the DermoScan clinical dermatology workstation, define clinical objectives, identify diagnostic pain points, and define system scope.

2. Summary of Work Done / Tasks Performed: Analyzed clinical challenges in dermatological screening (delayed specialist access, misdiagnosis of lookalike lesions like Tinea vs Eczema). Formulated system architecture: client-side privacy-first edge workstation, multi-quadrant clinical morphology assessment, deterministic heuristic risk engine, and ICD-10-CM crosswalk. Outlined technical stack: TypeScript 5.7, React 18.3, Vite 6, Tailwind CSS, Lucide React, HTML5 Canvas 2D API, and Gemini 2.5 Flash Vision API.

3. Skills/Concepts Learned: Clinical decision support systems (CDSS) architecture, HIPAA zero-telemetry edge processing principles, medical workflow optimization.

4. Challenges Faced: Scoping diagnostic boundary conditions to ensure the application acts as an adjunct triage tool rather than an autonomous medical diagnostic device.

5. Solutions/Support Received: Consulted project guidelines from Shaik. Arshad; incorporated mandatory disclaimer banners, non-definitive language, and emergency triage escalation protocols.

6. Conclusion/Reflection: Established solid project requirements, regulatory boundary conditions, and technical architecture for DermoScan.

4 August 2026 – Frontend Architecture, UI Layout & TypeScript Domain Schemas

1. Objectives of the Day: To scaffold the React + TypeScript application with Vite and design strong type definitions for dermatological data structures.

2. Summary of Work Done / Tasks Performed: Initialized Vite React-TS project with Tailwind CSS. Created comprehensive TypeScript interfaces in types.ts for LesionAnalysis, RiskLevel ('LOW' | 'MODERATE' | 'HIGH' | 'CRITICAL'), QuadrantAnalysis, DifferentialDiagnosis, TopicalPrecaution, and ICD10Code. Built the clinical workstation UI layout with responsive sidebars, image upload dropzone, analysis status badges, and tabs for Clinical Triage, ICD-10 Crosswalk, and Patient Education.

3. Skills/Concepts Learned: TypeScript strict type safety for medical payloads, discriminated unions, modular component architecture in React 18, Tailwind CSS utility-first styling.

4. Challenges Faced: Designing flexible schemas capable of representing primary, secondary, and tertiary differential diagnoses with varying confidence intervals.

5. Solutions/Support Received: Created normalized differential objects with dedicated fields for confidence score, diagnostic rationale, and contraindications.

6. Conclusion/Reflection: Successfully built a scalable UI foundation and type contracts for lesion analysis.

5 August 2026 – Client-Side Image Preprocessing & HTML5 Canvas Downsampling Engine

1. Objectives of the Day: To develop high-performance client-side image processing using the HTML5 Canvas 2D API for downsampling and privacy preservation.

2. Summary of Work Done / Tasks Performed: Implemented imageOptimizer.ts using HTML5 Canvas 2D bilinear interpolation. Built aspect-ratio-preserving constraint logic capping maximum dimensions at 1024x1024 pixels and converting uncompressed 15MB+ camera uploads into lightweight WebP/JPEG Base64 payloads (<250KB) in under 800ms. Added EXIF metadata stripping to preserve patient privacy on the client device prior to API transmission.

3. Skills/Concepts Learned: CanvasRenderingContext2D, bilinear downsampling algorithms, Base64 data encoding, client-side memory lifecycle, and offscreen canvas performance.

4. Challenges Faced: High-resolution mobile phone dermatological images (48MP+) caused memory spikes and UI freezing during image decoding.

5. Solutions/Support Received: Utilized asynchronous image onload callbacks and constrained canvas dimension math to avoid memory leaks and prevent main thread blocking.

6. Conclusion/Reflection: Achieved fast, client-side image optimization and privacy protection without server-side processing overhead.

6 August 2026 – Multimodal AI Vision Integration & Structured JSON Prompting

1. Objectives of the Day: To integrate Google Gemini 2.5 Flash Vision API and develop rigorous structured clinical prompts.

2. Summary of Work Done / Tasks Performed: Configured @google/genai client integration. Authored domain-specific dermatological system prompts forcing strict JSON output schema conforming to LesionAnalysis. Programmed prompts to extract primary lesion morphology (macule, papule, plaque, nodule, vesicle, bulla), secondary changes (scaling, crusting, lichenification), arrangement/distribution (linear, annular, clustered), and color spectrum (erythematous, violaceous, hyperpigmented).

3. Skills/Concepts Learned: Multimodal LLM prompting, vision transformer patch tokens, JSON mode constraints, zero-shot clinical feature extraction.

4. Challenges Faced: Vision models occasionally included conversational text or markdown code fences (```json), which broke standard JSON.parse().

5. Solutions/Support Received: Implemented a defensive sanitization pipeline with regex stripping (/```json\s*|\s*```/g) and fallback error boundaries before parsing.

6. Conclusion/Reflection: Secured deterministic and reliable JSON responses from the multimodal vision model.

7 August 2026 – Deterministic Heuristic Risk Engine & Multi-Factor Scoring

1. Objectives of the Day: To build a secondary deterministic heuristic risk evaluation engine independent of raw LLM confidence scores.

2. Summary of Work Done / Tasks Performed: Developed riskEngine.ts to calculate objective risk levels based on clinical dermatological criteria: lesion diameter (>6mm alert), asymmetry index, border irregularity, color variegation (ABCDE melanoma criteria), rapid evolution, ulceration, and anatomical risk zones (facial, acral, mucosal). Implemented weighted scoring matrix prioritizing red-flag morphological markers over benign features.

3. Skills/Concepts Learned: Rule-based clinical heuristic algorithms, risk stratification matrices, defensive AI validation, melanoma ABCDE criteria.

4. Challenges Faced: Calibrating heuristic weighting so that acute bacterial infections (e.g., cellulitis) trigger high/critical risk even when lesions lack pigment variegation.

5. Solutions/Support Received: Added dedicated infection heuristics evaluating rapid spreading border erythema, systemic warning flags, and warmth/tenderness indicators.

6. Conclusion/Reflection: Established a reliable two-tier safety validation mechanism combining AI vision outputs with deterministic clinical rules.

10 August 2026 – 4-Quadrant Clinical Morphology Engine & ICD-10-CM Crosswalk Module

1. Objectives of the Day: To implement the 4-quadrant morphological breakdown and map dermatological conditions to official ICD-10-CM billing and diagnostic codes.

2. Summary of Work Done / Tasks Performed: Developed the 4-Quadrant Morphology module dividing lesion assessment into Primary Morphology, Secondary Characteristics, Spatial Distribution, and Color Profile. Constructed the ICD-10-CM crosswalk dictionary covering Plaque Psoriasis (L40.0), Atopic Dermatitis (L20.9), Dyshidrotic Eczema (L30.1), Tinea Corporis (B35.4), Rosacea (L71.0), and Cellulitis (L03.9). Integrated differential diagnostic ranking with clinical rationale and contraindications (e.g., corticosteroid contraindication in Tinea infections).

3. Skills/Concepts Learned: Medical terminology standards, ICD-10-CM coding taxonomies, dermatological differential diagnosis logic, topical contraindication risks.

4. Challenges Faced: Accurately distinguishing clinical mimickers such as Tinea Incognito and contact dermatitis in the differential logic.

5. Solutions/Support Received: Enriched ICD-10 crosswalk entries with key discriminant clinical features (e.g., central clearing with active scaly border in fungal infections vs diffuse scaling in eczema).

6. Conclusion/Reflection: Enhanced clinical utility and diagnostic clarity for healthcare practitioners and trainees.

11 August 2026 – Non-Skin Anomaly Rejection Pipeline & Comprehensive Edge Testing

1. Objectives of the Day: To build a robust rejection pipeline for invalid images (non-skin images, low-lighting, blurry photos) and conduct cross-browser testing.

2. Summary of Work Done / Tasks Performed: Implemented automated image validation checks detecting non-cutaneous inputs (documents, objects, pets, landscapes) and image degradation (severe blur, extreme underexposure). Built user-friendly visual feedback dialogs advising optimal capture techniques (well-lit, perpendicular angle, sharp focus, neutral background). Conducted unit and integration testing across Chrome, Edge, Safari, and Firefox.

3. Skills/Concepts Learned: Edge-case validation, defensive prompt engineering, error state UX design, automated regression testing.

4. Challenges Faced: Handling edge cases where low-quality clinical photos caused the API to produce inconclusive or hallucinated results.

5. Solutions/Support Received: Added explicit confidence thresholds (<65% triggers 'Inconclusive / Retake Image' guidance) and strict non-skin rejection guardrails in the system prompt.

6. Conclusion/Reflection: Dramatically improved system robustness and eliminated false positive analyses on invalid image inputs.

12 August 2026 – Production Build Optimization, Vite Bundling & Vercel Edge Deployment

1. Objectives of the Day: To optimize the production bundle with Vite, configure environment variables, and deploy the application to Vercel Edge CDN.

2. Summary of Work Done / Tasks Performed: Ran npm run build to verify tree-shaking and TypeScript compilation. Configured vite.config.ts chunk splitting and asset optimization. Set up Git repository integration with GitHub (https://github.com/vijayesh26/Dermo-Scan-). Configured Vercel deployment pipeline (https://dermo-scan-nesn.vercel.app/) with automated CI/CD builds on push, HTTPS enforcement, and zero-telemetry client-side runtime environment.

3. Skills/Concepts Learned: Production Vite bundling, Vercel Edge Network architecture, CI/CD pipeline deployment, environment variable security (VITE_GEMINI_API_KEY).

4. Challenges Faced: Ensuring client-side API keys remain secure and avoiding unintended telemetry leakage in production logs.

5. Solutions/Support Received: Configured Vercel environment variables properly and ensured zero external logging of clinical images or patient metadata.

6. Conclusion/Reflection: Successfully deployed DermoScan live with fast sub-second global load times.

13 August 2026 – Clinical Documentation, Project Report Compilation & Slide Deck Finalization

1. Objectives of the Day: To complete comprehensive industrial training documentation, compile the final project report, and create presentation slides.

2. Summary of Work Done / Tasks Performed: Authored the 34-page formal Project Report (DERMO_SCAN_INDUSTRIAL_TRAINING_PROJECT_REPORT.docx and .pdf) following academic formatting guidelines. Incorporated real system screenshots (Clinical Triage, ICD-10 Crosswalk, Patient Education, Architecture). Compiled the 13-slide PowerPoint presentation (DERMO_SCAN_PROJECT_PRESENTATION.pptx) highlighting problem statement, tech stack, architecture, live demo links, and clinical impact. Rehearsed oral presentation flow.

3. Skills/Concepts Learned: Formal technical report writing, academic styling standards, presentation storytelling, clinical documentation.

4. Challenges Faced: Aligning complex technical architecture and medical terminology into simple, visually appealing presentation slides.

5. Solutions/Support Received: Refined presentation content with clean bullet points, visual cards, and clear diagnostic comparison tables.

6. Conclusion/Reflection: Completed all written documentation and presentation materials for university submission.

14 August 2026 – Capstone Project Demonstration, Viva Voce Defense & Internship Concluding Review

1. Objectives of the Day: To demonstrate the live DermoScan application to academic and industrial evaluation panels, defend technical choices during viva voce, and submit final deliverables.

2. Summary of Work Done / Tasks Performed: Delivered live demonstration of DermoScan running on Vercel (https://dermo-scan-nesn.vercel.app/). Showcased live image upload, client-side downsampling, real-time AI morphological analysis, ICD-10 mapping, and heuristic risk scoring. Defended technical design choices (React 18 vs Angular, Vite vs Webpack, client-side downsampling vs server-side preprocessing, defensive parsing, HIPAA zero-telemetry posture). Submitted project report, presentation, and daily diary.

3. Skills/Concepts Learned: Technical defense, viva voce communication, handling examiner queries on AI safety, model drift, and clinical liability.

4. Challenges Faced: Explaining edge-case handling when dermatological conditions have atypical clinical presentations.

5. Solutions/Support Received: Demonstrated the differential diagnosis matrix and highlighted the primary role of DermoScan as an adjunct triage tool rather than a final prescriber.

6. Conclusion/Reflection: Successfully completed the 8-week Industrial Training and Capstone Project with high appreciation from evaluators. Ready for engineering career in AI and Data Science.